

ASSIGNMENT-3

(a)

$$f(x) = e^{(x-1)^2} - 1 \Rightarrow f(1) = e^0 - 1 = 0$$

$$f'(x) = 2(x-1)e^{(x-1)^2} \Rightarrow f'(1) = 2(0)e^0 = 0$$

$$f''(x) = 2e^{(x-1)^2} [1 + 2(x-1)^2] \Rightarrow f''(1) = 2e^0 [1 + 0] = 2 \neq 0$$

The curve $f(x)$ touches the x -axis @ 1, without crossing it. $\rightarrow$ so, we can tell that we can have an even multiplicity.

$\rightarrow f'(x)$ curve is exactly zero @ $x=1$, $\Rightarrow$ means multiple root.

The $f''(x)$ curve has a value of 2 @ $x=1$

non-zero root

$$f(1) = 0, f'(1) = 0 \quad \text{f''(1) \neq 0} \quad \rightarrow \therefore x=1 \text{ is a double root}$$

with a multiplicity of 2

6

$$u(x) = \frac{f(x)}{f'(x)} = \frac{e^{(x-1)^2} - 1}{2(x-1)e^{(x-1)^2}}$$

$$\lim_{x \rightarrow 1} u(x) = \frac{f'(1)}{f''(1)} = \frac{0}{2} = 0$$

for a root of multiplicity m , the $u'(x)$

evaluated @ the roots $u'(x) = \frac{1}{m}$ $m=2$

$$u'(1) = \frac{1}{2} = 0.5 \neq 0$$

from plots;

$u(x)$ curve crosses the x -axis

transversally straight through @ $x=1$

with a non zero slope.

$\rightarrow u'(x)$ curve peaks @ 0.5 @ $x=1$

$$\therefore u(1)=0, u'(1) \neq 0$$

$\therefore u(x)$ has a root @ $x=1$ with

a multiplicity of 1.

© Standard secant method :-

$$x_{-1} = -0.5 \Rightarrow f(-0.5) = e^{\frac{(-0.5-1)^2}{-1}} = e^{\frac{2.25}{-1}} = 8.4877$$

$$x_0 = 0 \Rightarrow f(0) = e^{\frac{(0-1)^2}{-1}} = e^{-1} = 1.7183$$

1st iter:

$$x_1 = x_0 - f(x_0) \cdot \frac{x_0 - x_{-1}}{f(x_0) - f(x_{-1})}$$

$$x_1 = 0 - (1.7183) \left(\frac{0 - (-0.5)}{1.7183 - 8.4877} \right)$$

$$x_1 = -1.7183 \left(\frac{0.5}{-6.7694} \right) \approx 0.1269$$

2nd iter:

$$x_2 = x_1 - f(x_1) \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$$x_2 = 0.1269 - (1.1432) \left(\frac{0.1269 - 0}{1.1432 - 1.7183} \right)$$

$$x_2 \approx 0.3794$$

3rd iter.: $f(x_2) = f(0.3794) = e^{(0.3794-1)^2} - 1$

$= \approx 0.4697$

$$x_3 = x_2 - f(x_2) \left(\frac{x_2 - x_1}{f(x_2) - f(x_1)} \right)$$

$$x_3 = 0.3794 - (0.4697) \left(\frac{0.3794 - 0.1269}{0.4697 - 1.1432} \right)$$

$x_3 \approx 0.5554$

As we go on, Result
converges in 20 iterations to a root
 ≈ 0.999999 . [These are results
from the code].

-Results-

Standard Secant: root=0.999999, iterations=30, converged=True

Modified Secant: DID NOT CONVERGE (ran 63 iterations, drifted to x=-4.415e+13)

→ Modified Secant on $u(x)$:-

$$u(x) = \frac{1 - e^{-x^2}}{2x}$$

$$x_{-1} = -0.5 \Rightarrow u(-0.5) \approx -0.2982$$

$$x_0 = 0 \Rightarrow u(0) \approx -0.3161$$

It 1:-
$$x_1 = x_0 - u(x_0) \cdot \frac{x_0 - x_{-1}}{u(x_0) - u(x_{-1})}$$

$$\begin{aligned} x_1 &= 0 - (-0.3161) \frac{0 - (-0.5)}{-0.3161 - (-0.2982)} \\ &\approx -8.848 \end{aligned}$$

It 2:-

$$u(x_1) = u(-8.848) = \frac{1 - e^{-(8.848)^2}}{2(-8.848)} \approx -0.0508$$

$$x_2 = x_1 - u(x_1) \left(\frac{x_1 - x_0}{u(x_1) - u(x_0)} \right)$$

$$x_2 = -8.848 - (-0.0508) \left(\frac{-8.848}{-0.0508 - (-0.3161)} \right)$$

$$x_2 \approx -10.541$$

Third iter: (x_3):-

$$u(x_2) = u(-10.541) = \frac{-(-11.541)^2}{2(-11.541)} \approx -0.0433$$

$$x_3 = x_2 - u(x_2) \frac{x_2 - x_1}{u(x_2) - u(x_1)}$$

$$x_3 = -10.541 - (-0.0433) \left(\frac{-10.541 + 8.848}{-0.0433 + 0.0508} \right)$$

$$x_3 \approx -20.390$$

It diverges completely. The python code ran for 100 iterations and drifts to

$$x \approx -4.415 \times 10^3.$$

② Running using GEM, we got
me (x_1, x_2, x_3) as $(4, 8, -2)$.